

Improving Domain Generalization in Real-Time Semantic Segmentation through Rare-Class Tailored Techniques

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Abstract

This paper explores the application of Domain Adaptation (DA) techniques to semantic segmentation networks, aiming to address the domain shift problem between synthetic and real world data. An analysis of both standard and real time semantic segmentation architectures is conducted to evaluate their performance. Using a synthetic dataset as the source domain, the study assesses the degradation in model accuracy when tested on real world data. To improve generalization, data augmentation techniques tailored for rare classes—including a version of ClassMix—are applied. Furthermore, an adversarial domain adaptation approach is implemented to encourage the learning of domain invariant features. To specifically tackle the class imbalance problem, specialized loss functions are combined with the augmentation strategies. Additionally, a pretraining step focused on rare classes is employed to further enhance the model's ability to recognize underrepresented categories. This study provides insights into the effectiveness of combining pre-training, data augmentation, loss weighting, and adversarial training for real-time semantic segmentation under domain shift conditions. The code is available at https://github.com/AlessiaCicca/SemSeg_ML25.git.

1. Introduction

Semantic segmentation is a fundamental yet challenging task in computer vision that involves assigning a category label to every pixel in an image. Its goal is to partition an image into distinct regions representing different object categories, providing detailed pixel level understanding by accurately delineating object boundaries, unlike traditional object detection which only localizes objects [3]. However, models trained on one domain often struggle to perform well on different domains due to distribution shifts, motivating the use of Domain Adaptation (DA) methods. Domain

adaptation (DA) approaches aim to bridge the gap by aligning features and outputs between the source and target domains, thereby enhancing generalization. Rare classes often suffer from poor generalization, resulting in inaccurate predictions and fragmented segmentation masks. To address these limitations, this work investigates a multi stage training strategy focused on improving rare class segmentation under domain shift. In particular, our study is structured around:

- the implementation of a ClassMix variant tailored for rare classes, alongside the exploration of additional data augmentation techniques;
- the adoption of Adversarial Generative Domain Adaptation, with a comparative analysis between Weighted Cross Entropy Loss and Focal Loss as Segmentation Loss;
- the integration of Instance-aware Repeat Factor Sampling (IRFS) to rebalance rare class representation and the application of selective pixel masking during the early training.

2. Related Work

Semantic segmentation has seen major progress with the rise of deep learning, particularly through Fully Convolutional Networks (FCNs), which introduced pixel wise prediction. Among the influential models, DeepLab significantly advanced this task by incorporating atrous (dilated) convolutions, enabling wider receptive fields without losing spatial resolution [1]. DeepLabV2 further enhanced performance by using a ResNet101 backbone and introducing the Atrous Spatial Pyramid Pooling (ASPP) module, which captures multi scale contextual information through parallel atrous convolutions with different dilation rates. This architecture achieved strong accuracy on benchmark datasets but remained computationally intensive, limiting its use in real time applications [1].

To address this limitation, BiSeNet (Bilateral Segmentation Network) was proposed as a lightweight and fast alternative. It uses a two path structure: a Spatial Path to keep detailed spatial information, and a Context Path to efficiently capture semantic features using a compact backbone. To refine these features, BiSeNet incorporates Attention Refinement Modules (ARM) and merges the two branches via a Feature Fusion Module (FFM) that adaptively balances spatial and contextual features. This design achieves a favorable trade off between speed and accuracy, making it suitable for real time scenarios [10].

3. Datasets

Two datasets are employed in this work. Both datasets are designed for semantic urban scene understanding and contain 19 semantic classes targeted for segmentation. However, they both suffer from class imbalance, with some categories being significantly underrepresented, as illustrated in Figure 1.

- **Cityscape:** Cityscapes is a high quality dataset consisting of manually annotated street view images captured across multiple cities under diverse weather and lighting conditions [2]. In this work, in the initial phase, 1,572 images are used for training and 500 for validation of the DeepLabV2 and BiSeNet models. Under the domain adaptation setting, Cityscapes is utilized exclusively for evaluation as target domain.
- **GTA:** The GTA5 dataset is introduced to perform domain adaptation. It is a large scale synthetic dataset generated from the video game Grand Theft Auto V. It enables rapid generation of pixel level labeling, that is over 500 times faster than manual annotation [7]. This makes GTA5 a cost effective and scalable resource for domain adaptation and other computer vision applications. In this study, a limited subset is employed as source domain, comprising 1,750 images for training and 750 for validation.

4. Preliminary Work

The study begins by establishing foundational baselines through the independent training of two widely recognized semantic segmentation models: DeepLabV2 and BiSeNet. These models are evaluated on the Cityscapes dataset under standard supervised learning conditions. This step serves as a crucial reference point, enabling a clear understanding of each model’s performance.

Initially, the **DeepLabV2** architecture was trained on the Cityscapes dataset. After extensive experimentation with various configurations, the set of architectural components and hyperparameters that yielded the best performance is summarized in Tab. 1.

Table 1. Hyperparameters and architectural components for DeepLabV2 and BiSeNet configurations used in this study.

Component	DeepLabV2	BiSeNet
Backbone	ResNet101	ResNet18
Pretraining	ImageNet	ImageNet
Loss Function	CrossEntropy	CrossEntropy
Optimizer	SGD + momentum	SGD + momentum
Learning Rate	0.001	0.01
Momentum	0.9	0.9
Weight Decay	0.0005	0.0005
Batch Size	4	4
Epochs	50	50

Table 2. Performance and complexity comparison of DeepLabV2 and BiSeNet on the Cityscapes Validation Set.

Metric / Model	DeepLabV2 (R101)	BiSeNet (ResNet-18)
mIoU (%)	47.40	48.45
Latency (ms)	129.93	16.56
FLOPs (G)	374.66	25.90
Parameters (M)	43.90	12.58

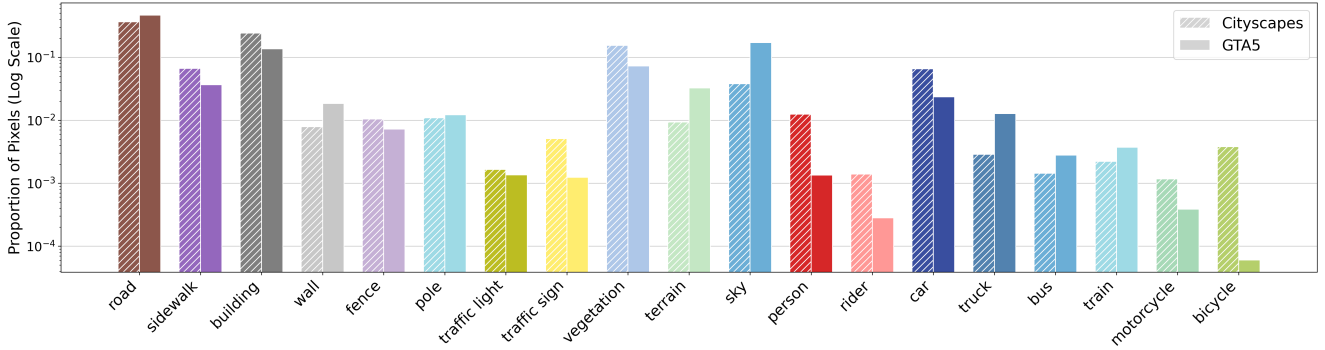
Subsequently, the **BiSeNet** architecture, distinguished by its design for real time semantic segmentation, was also subjected to training on the Cityscapes dataset. Its configuration was designed to achieve a balance between inference speed and segmentation accuracy. The configuration that delivered the highest performance is presented in Tab. 1.

Following these individual training regimens, a comprehensive comparative analysis was conducted between the DeepLabV2 and BiSeNet architectures. Tab. 2 presents a detailed overview of their respective performance and computational complexities as observed on the Cityscapes evaluation set. This comparison is quantitatively articulated through several key metrics:

- **Mean Intersection over Union (mIoU):** Measures the average overlap between predicted segmentation masks and ground truth across all semantic classes;
- **Latency:** Refers to the time taken by the model to process a single image during inference;
- **FLOPs (Giga Floating Point Operations):** Represents the number of floating-point operations required for a single forward pass;
- **Parameters:** Indicates the total number of learnable weights in the model.

BiSeNet demonstrates an effective balance between accuracy and computational efficiency, slightly outperforming DeepLabV2 in terms of mIoU (48.45% vs. 47.40%), while significantly reducing latency (16.56 ms vs. 129.93 ms),

Figure 1. Normalized pixel distribution per semantic Class: Cityscapes vs. GTA5. This bar chart presents the proportion of pixels for the classes within the Cityscapes (train-subset) and GTA5 datasets used in this study. Each class is represented by a unique color, and the datasets are distinguished by their fill patterns, according to the legend.



FLOPs (25.90 G vs. 374.66 G), and parameter count (12.58 M vs. 43.90 M). These advantages come from its dual path architecture and lightweight backbone, which enable real time inference with only a marginal compromise in accuracy. In contrast, DeepLabV2 leverages a deeper backbone and the ASPP module to prioritize segmentation fidelity, at the cost of increased computational demand. Based on these preliminary insights and according to the requirements, BiSeNet was selected for the subsequent stages of this work.

5. Methods

In this section, we delineate the various methodological approaches employed to address the challenges of semantic segmentation in domain adaptation scenarios, with a special focus on improving the model’s performance on rare classes.

5.1. Augmentation

Data augmentation is a technique used in machine learning to increase the quantity and diversity of training data. This is crucial when data availability is limited, as a larger and more varied dataset helps the model generalize better and reduce overfitting. [5]

We conducted a series of experiments to evaluate the impact of various data augmentation strategies on the performance of our semantic segmentation model when applied to the Cityscapes dataset. This investigation aimed to discern which augmentation configurations contribute most effectively to model generalization.

The configuration of techniques we employ includes the following:

- **Random crop + Horizontal flip:** This strategy employs random cropping and horizontal flipping to increase spatial variability. Cropping extracts fixed size

regions, resizing smaller inputs if needed, while flipping mirrors image-mask pairs with 0.5 probability, promoting spatial invariance and contextual diversity.

- **Horizontal flip:** In this case, only horizontal flipping is applied, introducing left right invariance without additional augmentations.
- **Color jitter:** Photometric augmentation that randomly changes color properties— gamma, saturation, hue, brightness, and contrast—in different orders to improve robustness against lighting and camera variations.
- **Set A:** This set combines several geometric and photometric transformations to provide a robust general purpose augmentation scheme.
- **Set B:** This augmentation strategy applies Horizontal Flip, Random Scaling, Random Crop, and Color Jitter transformations, along with a version of ClassMix technique, which will be described later.

The comprehensive performance of the models trained with different data augmentation strategies is summarized in Table 3. This wide table presents the overall **mIoU** alongside detailed per class mIoU scores across all 19 Cityscapes categories. The initial row, labeled ‘None’, serves as a crucial baseline. It reflects the performance of the BiSeNet model when trained on the synthetic GTA5 source domain and evaluated on the Cityscapes target domain, without the application of any additional data augmentation techniques. To clarify the specific combinations of these augmentation methods, Table 4 offers a clear summary, listing the individual data augmentation techniques used in each tested strategy (A and B).

Table 3. Comparison of different Data Augmentation strategies and mIoU per Class on Cityscapes.

Method	mIoU	road	sidewalk	building	wall	fence	pole	traffic light	traffic sign	vegetation	terrain	sky	person	rider	car	truck	bus	train	motorcycle	bicycle
None	0.121	0.282	0.129	0.445	0.009	0.032	0.000	0.000	0.010	0.494	0.006	0.254	0.000	0.000	0.443	0.186	0.000	0.005	0.000	0.000
Random Crop + Horizontal Flip	0.149	0.566	0.121	0.613	0.042	0.033	0.062	0.000	0.000	0.236	0.040	0.700	0.018	0.000	0.373	0.023	0.003	0.000	0.000	0.000
Horizontal Flip	0.181	0.548	0.094	0.612	0.080	0.070	0.165	0.067	0.049	0.637	0.035	0.686	0.106	0.000	0.455	0.057	0.003	0.000	0.000	0.000
Color Jitter	0.227	0.671	0.212	0.669	0.106	0.084	0.235	0.065	0.038	0.704	0.163	0.751	0.059	0.011	0.458	0.070	0.015	0.025	0.006	0.000
Set A	0.210	0.612	0.288	0.576	0.072	0.053	0.163	0.033	0.026	0.698	0.177	0.760	0.106	0.000	0.413	0.011	0.080	0.000	0.000	0.000
Set B	0.160	0.374	0.178	0.322	0.094	0.050	0.156	0.041	0.024	0.629	0.074	0.563	0.228	0.000	0.276	0.025	0.000	0.000	0.010	0.000

Table 4. Definition of Augmentations Strategies A and B

	Horizontal Flip	Random Scaling	Random Crop	Brightness Adjustment	Hue Shift	Gamma Correction	Saturation Adjustment	Contrast Adjustment	Color Jitter	ClassMix
A	✓	✓	✓	✓	✓	✓	✓	✓	-	-
B	✓	✓	✓	-	-	-	-	-	✓	✓

5.1.1 ClassMix

To directly address the problem of class imbalance we integrated a targeted variant of ClassMix into our data augmentation pipeline. This technique was originally proposed by Olsson *et al.* [5]. ClassMix is an advanced data augmentation technique designed specifically for semantic segmentation. Unlike simpler transformations, ClassMix does not merely modify an image but composes it with another by focusing on semantically meaningful regions.

Given a primary image A with its segmentation ground truth S_A , and a secondary image B with its ground truth S_B , ClassMix generates a new composite image X_A and its corresponding ground truth Y_A .

The original ClassMix method typically involves selecting a random class and mixing segments based on that class from one image to another. However, in our specific implementation, designed to mitigate class imbalance, we introduce a key modification: our approach prioritizes including content from rare classes present in the secondary image (B) onto the primary image (A).

The augmented method operates according to the following sequence:

1. **Rare Class Image Selection:** Our method first identifies an image B from the dataset that is guaranteed to contain at least one of the classes — we selected the classes traffic light, person and motorcycle, in accordance with the distribution shown in Figure 1. The primary image A is processed from the training batch and does not necessarily contain rare classes.
2. **Rare Class Mask Creation:** A binary mask M is generated based exclusively on the identified rare classes present in image B . The mask M has a value of 1 (true) for all pixels belonging to any rare class in S_B ,

and 0 (false) otherwise.

$$M(p) = \begin{cases} 1 & \text{if } S_B(p) \in \text{RareClasses} \\ 0 & \text{otherwise} \end{cases}$$

3. **Image Combination:** The composite image X_A is constructed by selectively combining pixels from A and B according to the mask M :

$$X_A(p) = M(p) \cdot B(p) + (1 - M(p)) \cdot A(p) \quad (1)$$

Similarly, the new ground truth segmentation map Y_A is formed by combining S_A and S_B using the same mask M .

Figure 2 shows a schematic visualization of our ClassMix variant for rare class augmentation.

This process enriches the training data with challenging examples of underrepresented categories, aiming to improve segmentation accuracy for rare classes and effectively address the class imbalance problem.

5.2. Adversarial Generative Domain Adaptation

Manually labeling diverse target domains is labor intensive and costly. Unsupervised domain adaptation addresses this by training models on labeled synthetic datasets and adapting them to unlabeled real world domains, enabling accurate segmentation without requiring target annotations.

Among the various techniques for this task, adversarial generative methods are widely employed in domain adaptation to minimize the discrepancy between source and target domains. This framework fundamentally consists of a feature generator and a domain discriminator trained adversarially. The generator G , aims to extract features that are indistinguishable across domains, while the discriminator D learns to distinguish whether the extracted features originate from the source or the target domain. Through

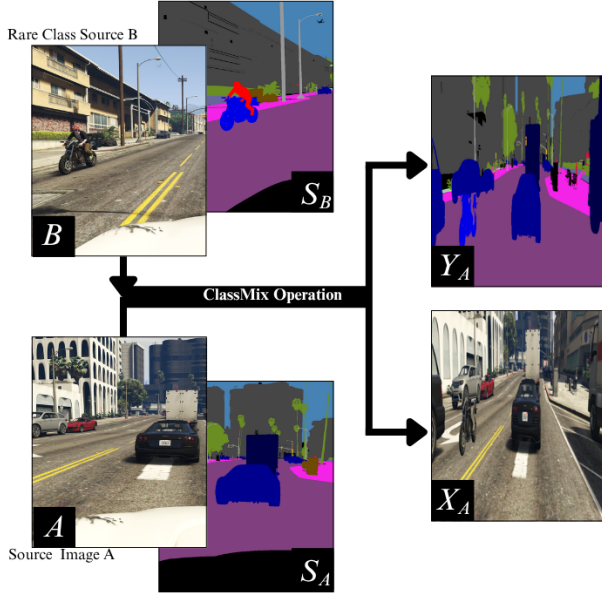


Figure 2. Schematic visualization of our ClassMix variant for rare class augmentation.

this adversarial training, the generator is incentivized to produce domain invariant features that confuse the discriminator, thereby enhancing the model’s ability to generalize to the target domain. This makes adversarial generative methods a powerful and scalable solution for real world domain adaptation scenarios. [11] [8]

5.2.1 Implementation

Our setup employs BiSeNet with a ResNet-18 backbone as segmentation network, which serves as the generator G and is trained to predict pixel wise class labels. This is the same architecture adopted in the earlier stages of our work. To perform domain adaptation, we introduce a fully convolutional discriminator D, consisting of five 4x4 convolutional layers with leaky ReLU activations and dropout for regularization. The discriminator operates directly on the softmax outputs of G, learning to distinguish between source and target domain predictions.

Data preparation involves using the GTA5 dataset as the source domain, which provides pixel-level annotations, and the Cityscapes dataset as the unlabeled target domain for adaptation. Input images are subjected to data augmentation techniques for the source domain to enhance robustness.

5.2.2 Objective function

The total objective function for the adaptation task combines these two losses:

$$\mathcal{L}(I_s, I_t) = \mathcal{L}_{seg} + \lambda_{adv} \mathcal{L}_{adv} \quad (2)$$

Here, I_s denotes a source image, and I_t denotes a target image. $\mathcal{L}_{seg}$ is the supervised segmentation loss, $\mathcal{L}_{adv}$ is the adversarial loss. The hyperparameter λ_{adv} balances the influence of the segmentation loss and the adversarial loss.

The optimization of the network explicitly follows a min max criterion:

$$\max_D \min_G \mathcal{L}(I_s, I_t) \quad (3)$$

We want to minimize the segmentation loss within the generator for images originating from the source domain, while simultaneously maximizing the probability that predictions for target domain images are classified as if they originated from the source domain. This adversarial setup compels the generator to produce target segmentation that are perceptually consistent with the source domain’s characteristics, thereby bridging the domain gap. [8]

5.3. Loss-Based Adaptation Strategies

The choice and configuration of loss functions play a pivotal role in training robust semantic segmentation models, especially when dealing with class imbalance and domain shift.

For the supervised segmentation task of the generator network, we investigated two distinct loss functions. These losses aim to optimize the pixel wise classification performance where ground truth labels are available.

- **Weighted Cross Entropy Loss (Weighted CE):** To specifically address the adverse effects of class imbalance, where common classes often dominate the loss landscape, we extended the standard Cross Entropy Loss by incorporating class specific weights. These weights, denoted as w_c , are applied to each class’s contribution to the loss, effectively giving higher importance to rare classes during training. The formula for the weighted CE Loss is:

$$\mathcal{L}_{\text{Weighted CE}} = - \sum_{p \in P} \sum_{c=1}^C w_c y_{p,c} \log(\hat{y}_{p,c}) \quad (4)$$

In our approach, the weights for each class were calculated based on inverse pixel frequencies following the scheme proposed by Paszke et al. [6]:

$$w_c = 1.0 / \log(1.02 + f_c) \quad (5)$$

where w_c is the weight for class c and f_c is the pixel frequency of class c in the training dataset.

- **Focal Loss:** Introduced by Lin et al. [4], Focal Loss was designed to mitigate the class imbalance problem by down weighting the contribution of well classified examples. This mechanism redirects the training focus towards hard, misclassified examples. The formula for Focal Loss is defined as follows:

$$L_{Focal} = - \sum_{i=1}^N \sum_{c=1}^C (1 - \tilde{y}_{i,c})^\gamma y_{i,c} \log(\tilde{y}_{i,c}) \quad (6)$$

where $\gamma \geq 0$ is a focusing parameter. Similar to Weighted CE, class weights calculated as $w_c = 1.0 / \log(1.02 + f_c)$ were also applied within the Focal Loss formulation to further emphasize rare classes. In line with standard practices to improve comparability and stability during training, the loss was averaged over the number of examples.

For the domain adaptation component, a Binary Cross-Entropy (BCE) loss is used by the discriminator to distinguish between features originating from the source and target domains, and by the generator to produce features that are indistinguishable from the source domain.

- **Binary Cross Entropy Loss (BCE):** This loss is fundamentally used for binary classification tasks. In our adversarial framework, it is applied to guide the training of both the discriminator and the generator’s adversarial component. The discriminator aims to classify inputs as either from the source domain ($y = 1$) or the target domain ($y = 0$), while the generator aims to produce target domain outputs that fool the discriminator into classifying them as source domain ($y = 1$). The formula for BCE Loss for a single sample is:

$$\mathcal{L}_{BCE}(y, p) = -[y \log(p) + (1 - y) \log(1 - p)] \quad (7)$$

where y is the true binary label (0 or 1) and p is the predicted probability for the positive class.

5.4. Pretraining Strategy for Rare Class

According to the aim of our extension, we propose a pretraining strategy aimed at improving the model’s ability to recognize and segment rare classes. These are identified by analyzing class frequencies across the dataset and selecting those present in less than a given percentage of images (e.g., 20%). To address the imbalanced nature of the dataset, we employ **Instance-Aware Repeat Factor Sampling (IRFS)** throughout the entire pretraining stage. This strategy, inspired by Yaman et al. [9], increases the sampling frequency of images containing rare classes based on a computed repeat factor, which takes into account both image frequency and instance frequency.

Let $f(i, c)$ be the fraction of images containing at least one instance of class c , and $f(b, c)$ be the fraction of instances of class c in the dataset. We define the category-level frequency f_c as the geometric mean of these two quantities. Then, the category level repeat factor r_c is computed as:

$$r_c = \max \left(1, \sqrt{t / \sqrt{f(i, c) \times f(b, c)}} \right) \quad (8)$$

where t is a threshold hyperparameter. Classes with $f_c \geq t$ are not oversampled (i.e., $r_c = 1$). The image-level repeat factor r_i for each image i is then defined as $r_i = \max_{c \in i} r_c$, where $c \in i$ denotes the set of classes present in image i . Each image is repeated $\lceil r_i \rceil$ times in the training dataset. This instance-aware sampling strategy balances the dataset more effectively by prioritizing images containing truly rare classes, both in terms of image presence and instance counts, leading to better representation learning and improved segmentation performance.

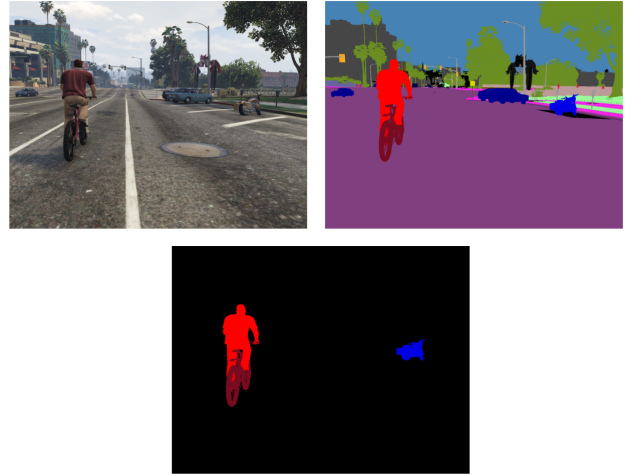


Figure 3. The top-left panel shows the original image, the top-right panel presents a semantic segmentation mask of the original image, while the bottom panel displays a specific masking result, highlighting only *bicycle*, *rider*, and *motorcycle* classes.

During the entire pretraining, IRFS with a Weighted Cross Entropy Loss is applied to increase exposure to rare classes. Additionally, in the first six epochs, a selective pixel masking is applied where all pixels not belonging to rare classes are set to `ignore_index` (255), so they do not contribute to the loss, as shown in Fig. 3. This encourages the model to focus exclusively on rare classes early on, while still benefiting from IRFS sampling. In this early phase, a standard Cross Entropy Loss is used without weighting since only rare classes are considered. This two step training procedure encourages the model to learn stronger and more discriminative features for rare classes.

Table 5. Comparative Performance of Domain Adaptation Strategies on Cityscapes Validation Set (mIoU per Class).

Method	mIoU	road	sidewalk	building	wall	fence	pole	traffic light	traffic sign	vegetation	terrain	sky	person	rider	car	truck	bus	train	motorcycle	bicycle
Best Augmentation	0.227	0.671	0.212	0.669	0.106	0.084	0.235	0.065	0.038	0.704	0.163	0.751	0.059	0.011	0.458	0.070	0.015	0.025	0.006	0.000
DA with Cross-Entropy	0.163	0.632	0.157	0.546	0.056	0.001	0.043	0.000	0.001	0.446	0.083	0.617	0.000	0.000	0.449	0.038	0.015	0.000	0.005	0.001
DA with Focal Loss	0.220	0.671	0.232	0.591	0.053	0.129	0.107	0.071	0.109	0.411	0.055	0.640	0.309	0.045	0.574	0.069	0.082	0.004	0.021	0.010
Pretrain + DA with Focal Loss (15+35)	0.240	0.678	0.317	0.613	0.076	0.116	0.165	0.153	0.108	0.639	0.095	0.644	0.295	0.090	0.461	0.045	0.023	0.003	0.029	0.016
Pretrain + DA with Focal Loss (15+50)	0.243	0.673	0.325	0.624	0.075	0.119	0.168	0.168	0.112	0.638	0.108	0.642	0.303	0.090	0.449	0.043	0.039	0.005	0.025	0.021

6. Experiments and Results

For the augmentation phase without Domain Adaptation, we employed the architecture presented in Table 1 with a weighted Cross Entropy Loss, where the class weights were computed according to Equation (5). This weighting scheme aims to give higher importance to rare classes, thereby improving the model’s ability to correctly classify underrepresented categories.

The table Sec. 5.1 presents the mean Intersection over Union (mIoU) and class wise IoU scores for different data augmentation methods applied during training. All the augmentation techniques improve performance compared to the baseline with no augmentation (*None*), which achieves the lowest mIoU of 0.121. Among the augmentations, *Color Jitter* stands out as the most effective method, achieving the highest overall mIoU score of 0.227, outperforming other augmentations such as *Random Crop + Horizontal Flip* (0.149), *Horizontal Flip* (0.181), *Set A* (0.210), and *Set B* (0.160). This indicates that color variation in training images significantly enhances the model’s ability to generalize. Although *Set B*, which applies class mix augmentation specifically to *persons*, *traffic lights*, and *motorcycles*, does not surpass the mIoU of *Color Jitter*, it provides notable improvements for these particular classes. For instance, the IoU for *persons* increases to 0.228 (compared to 0.059 for *Color Jitter*), traffic light improves to 0.041 (vs. 0.065), and motorcycle reaches 0.010 (vs. 0.006). This suggests that the targeted class mix strategy in *Set B* benefits these specific categories despite the lower overall mIoU and that other combinations using this approach could potentially lead to further improvements.

As indicated, since *Color Jitter* proved to be the most effective augmentation method, it was adopted in the subsequent training steps of Domain Adaptation. The architecture and hyperparameters were implemented following the setup described in [8], with the learning rate adjusted to 2.5×10^{-5} after testing revealed it to be the optimal value. Specifically, a polynomial decay scheduler was used for both the main optimizer and the discriminator optimizer. This scheduler gradually reduces the learning rate as training progresses, helping to stabilize the optimization by decreasing the step size over the epochs. Modifications were also introduced to the discriminator architecture by reduc-

ing the number of base filters from 64 to 32 and adding dropout layers. These changes were motivated by computational constraints and aimed at improving generalization, as the original architecture tended to produce a strong discriminator that hindered stable training. The loss functions employed during training are detailed in Section 5.3. Although the results obtained using the best performing augmentation method are superior in terms of overall mIoU, the Domain Adaptation (DA) approach with Focal Loss demonstrates significant improvements specifically for rare classes, according to Figure 1. This method substantially increases

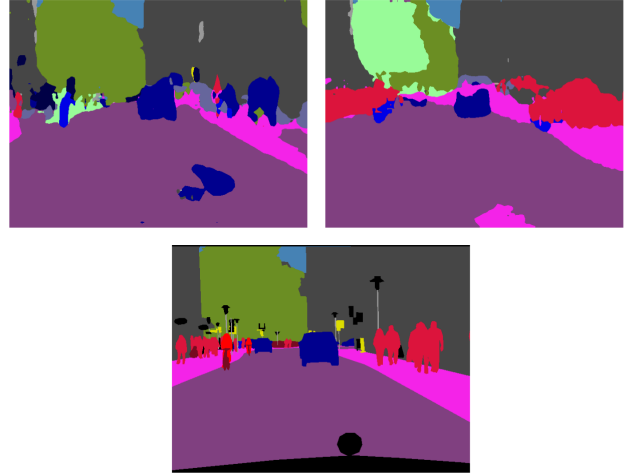


Figure 4. Comparison of semantic segmentation results. Top-left: prediction without Domain Adaptation (D.A.) using Color Jitter, top-right: prediction with Domain Adaptation (D.A.) using Focal Loss, bottom: ground truth.

the IoU for underrepresented categories such as *traffic sign*, *person*, and *bus*, as can be seen in Fig. 4 and Tab. 5. This aligns with the well established effectiveness of Focal Loss in addressing class imbalance by reducing the contribution of well classified examples and emphasizing harder, misclassified instances. Given these findings, it is reasonable to hypothesize that extending the training over a larger number of epochs or adjusting the γ parameter in the Focal Loss formulation could lead to significantly improved performance.

Such refinements may further enhance the model’s ability to learn from challenging classes and potentially exceed the current best results. By contrast, the use of weighted Cross-Entropy does not yield satisfactory results. Not only does this method perform worse than the best data augmentation strategies in terms of overall mIoU, but it also fails to provide notable improvements for rare classes.

The pretraining was performed following the strategy described in 5.4. Subsequently, additional training was carried out for 35 and 50 epochs using the domain adaptation architecture illustrated above, where a γ parameter equal to 1.2 was used, based on earlier observations. This procedure allowed us to achieve the best combination reported in this work, according to Tab. 5, and its result will be discussed in the following section.

7. Discussions and Findings

Future work could explore more sophisticated data augmentation techniques to enrich the training data, as all our current attempts resulted in worse performance compared to a simple Color Jitter. Additionally, the selection of rare classes for ClassMix should be automated, as manually specifying them is not a robust or scalable approach. Moreover, additional training cycles with varied hyperparameters could help improve the model’s robustness to domain shifts. These directions could potentially lead to further performance gains and a more stable and generalizable training process.

Overall, the techniques we adopted proved to be effective, as they allowed us to significantly improve segmentation predictions, particularly for rare and underrepresented classes. These results confirm the validity of our proposed pretraining strategy and domain adaptation framework. Comparing the best configuration in our pipeline with the baseline model of training on GTA5 and evaluating on Cityscapes, we observe substantial improvements across nearly all semantic classes.

The overall mIoU increases from **12,1 %** to **24,3 %**, effectively doubling the segmentation quality.

Some of the most significant per-class improvements include:

- **Road:** from 0.282 to 0.673 ($\Delta = +0.391$)
- **Person:** from 0.000 to 0.303 ($\Delta = +0.303$)
- **Traffic light:** from 0.000 to 0.168 ($\Delta = +0.168$)

Furthermore, rare classes that were completely undetected in the baseline (such as *rider*, *traffic light*, and *motorcycle*) are now predicted with non zero accuracy, demonstrating the effectiveness of the rare class focused pretraining and the use of focal loss in handling class imbalance.

This confirms that our method not only boosts general performance, but also significantly enhances prediction quality for underrepresented categories.

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